

Risk Analysis Report

Software Testing Final Report

Course: Software Testing

Task Management Platform

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Contents

1	Project Name	3
2	Project Overview	3
2.1	Application Background	3
2.2	Functional Scope	3
2.3	Technical Background	3
3	Risk Analysis Method	4
3.1	Analysis Approach	4
3.2	Risk Level Definition	4
3.3	Assessment Dimensions	5
4	Requirement-Level Risk Analysis	5
4.1	Overall Result	5
4.2	Requirement Analysis	6
4.2.1	REQ-001 User Registration	6
4.2.2	REQ-002 User Login	7
4.2.3	REQ-003 Password Reset	7
4.2.4	REQ-004 Project Management	8
4.2.5	REQ-005 Task Management	8
4.2.6	REQ-006 Task Assignment	8
4.2.7	REQ-007 Task Statistics	9
5	Technical Risk Analysis	9
5.1	Security Risks	9
5.2	Data Consistency Risks	10
5.3	Testability Risks	10
6	Test Arrangement Based on Risks	10

6.1	Test Priority Allocation	10
6.2	Testing Resource Suggestions	11
7	Conclusion	11

1 Project Name

Task Management Platform

2 Project Overview

2.1 Application Background

The system under test is a task management platform implemented with Flask and SQLite. It is designed for task collaboration in small teams. The system provides three main modules: user authentication, project management, and task management. It exposes its services through RESTful APIs, while the backend code is organized by routes, data models, and validation logic.

The purpose of this report is to identify and classify the major risks of the target application before the testing activities start. The result is used to guide the later test plan, testing effort allocation, and selection of testing techniques.

2.2 Functional Scope

Module	Main Functions	Main APIs
User Authentication	Registration, login, password reset, logout	/api/register, /api/login, /api/forgot-password, /api/reset-password, /api/logout
Project Management	Project creation, query, update, deletion, statistics	/api/projects, /api/projects/<id>, /api/projects/<id>/stats
Task Management	Task creation, update, deletion, assignment, query by user	/api/projects/<id>/tasks, /api/tasks/<id>, /api/tasks/<id>/assign, /api/users/<name>/tasks

2.3 Technical Background

Dimension	Content
Programming Language	Python 3.11+
Web Framework	Flask 3.0
Data Storage	SQLite
Architecture	Blueprint-based route separation and factory pattern
Test Execution	Pytest with Flask <code>test_client()</code>

3 Risk Analysis Method

3.1 Analysis Approach

This report follows a "requirement decomposition + risk assessment + manual review" approach. The main steps are:

1. Decompose the system into seven standard requirements according to the requirement document.
2. Assess risk from functional criticality, data sensitivity, security impact, business complexity, and user visibility.
3. Assign a risk level and testing priority to each requirement.
4. Add technical risk judgment based on the implementation of the target application.

3.2 Risk Level Definition

Risk Level	Definition
High	Involves authentication, security, sensitive data, or core business flows. A failure would directly affect system availability or security.
Medium	Involves major business functions. A failure would affect user operations or data consistency, but may not immediately break core authentication.
Low	Auxiliary or display-oriented function. A failure has a smaller impact and usually does not block core business use.

3.3 Assessment Dimensions

Dimension	Focus
Functional Criticality	Whether the requirement belongs to a core business path
Data Sensitivity	Whether passwords, accounts, or personal information are involved
Security Impact	Whether authentication bypass or unauthorized access may be introduced
Business Complexity	Whether many condition combinations, state transitions, or branches are involved
User Visibility	Whether the problem is directly visible to end users

4 Requirement-Level Risk Analysis

4.1 Overall Result

Requirement					Main Risk
ID	Function	Risk Score	Risk Level	Test Priority	Factors
REQ-001	User registration	90	High	High	Many input fields, complex password rules, user uniqueness
REQ-002	User login	92	High	High	Locking logic, failure counting, authentication correctness

Requirement					Main Risk
ID	Function	Risk Score	Risk Level	Test Priority	Factors
REQ-003	Password reset	88	High	High	Token validity, password update, consistency of security messages
REQ-004	Project management	68	Medium	Medium	Data integrity after project deletion, name validation
REQ-005	Task management	64	Medium	Medium	Task field validation, status and priority constraints, date handling
REQ-006	Task assignment	66	Medium	Medium	User existence validation, maintenance of task-user relationships
REQ-007	Task statistics	35	Low	Low	Accuracy of statistics, read-only aggregation logic

4.2 Requirement Analysis

4.2.1 REQ-001 User Registration

This requirement contains several input fields, including username, password, password confirmation, email, and age. It involves format validation, boundary handling, and database uniqueness checks. Its result directly affects account creation and is the first entry point for users, so

the risk level is set to High.

Main risks include:

- Weak passwords may be accepted incorrectly.
- Duplicate usernames or emails may not be blocked correctly.
- Age boundary values may be handled inconsistently.
- Unstable validation order may lead to inconsistent responses.

The suggested testing focus is equivalence partitioning, boundary value analysis, and condition-combination testing.

4.2.2 REQ-002 User Login

The login function involves not only username and password matching, but also failure counting and account locking logic. Because it directly affects user authentication and contains clear state changes, it is classified as a high-risk requirement.

Main risks include:

- Consecutive failed login attempts may be counted incorrectly.
- Failure records may not be cleared correctly after successful login.
- Lock duration may be calculated incorrectly.
- Locking and unlocking conditions may be unstable.

The suggested testing focus is decision table testing, state transition testing, and boundary value analysis.

4.2.3 REQ-003 Password Reset

Password reset is related to account recovery and secondary authentication security. It includes token generation, expiration validation, and new password constraints. If implemented incorrectly, it may prevent legitimate users from recovering accounts or introduce clear security risks. Therefore, the risk level is High.

Main risks include:

- Token expiration time may be calculated incorrectly.
- Invalid or expired tokens may not be recognized correctly.
- New password validation may be insufficient.
- Prompt messages may reveal account status.

The suggested testing focus is token state coverage, time boundary testing, and security consistency checks.

4.2.4 REQ-004 Project Management

Project management includes project creation, update, deletion, and statistics. It is the upper-level container for task management. Although it does not directly involve account security, data consistency after project deletion and project name validity affect later task operations. Therefore, the risk level is Medium.

Main risks include:

- Orphan tasks may remain after a project is deleted.
- Empty, overly long, or duplicate project names may be handled incorrectly.
- API responses may be inconsistent when a project does not exist.

The suggested testing focus is boundary value analysis, integration testing, and data consistency verification after deletion.

4.2.5 REQ-005 Task Management

Task management covers core operations such as creating, updating, deleting, and viewing tasks. It includes several business fields, such as title, priority, status, and due date. In this project, the risk level is set to Medium rather than Low because this module is one of the most frequently used business modules and has many internal constraints.

Main risks include:

- Illegal priority or status values may be accepted incorrectly.
- Due date validation may be incomplete.
- Tasks may still be created when the project does not exist.
- The handling rules for completed tasks may not match the requirement.

The suggested testing focus is equivalence partitioning, boundary value analysis, decision table testing, and path coverage.

4.2.6 REQ-006 Task Assignment

Task assignment connects task objects and user objects, and is a key part of collaboration. It must ensure consistency between assignee input and user information. Therefore, the risk level is Medium.

Main risks include:

- Non-existent assignees may be accepted incorrectly.
- Query APIs may not reflect assignment results correctly.
- The assignee field may lack constraints during task update.

The suggested testing focus is equivalence partitioning, decision table testing, and state transition testing.

4.2.7 REQ-007 Task Statistics

Task statistics is mainly a read-only query function. Its business complexity and input complexity are relatively low, but the aggregation range and statistical results still need to be correct. Therefore, the risk level is Low.

Main risks include:

- The statistical scope may be wrong and include unrelated tasks from other projects.
- The statistical rules may be inconsistent.
- The system may not return correct errors when a project does not exist.

The suggested testing focus is decision table testing, path coverage, and data-driven testing.

5 Technical Risk Analysis

5.1 Security Risks

Risk Item	Level	Description
Limited password hash strength	High	The current implementation uses SHA-256 and lacks a stronger dedicated password hashing mechanism.
Unstable login failure record handling	High	Login and locking logic depend on state and may easily cause incorrect locking.

Risk Item	Level	Description
Password reset validity control	High	Token expiration judgment and time handling are related to account security.

5.2 Data Consistency Risks

Risk Item	Level	Description
Cascading handling after project deletion	Medium	If tasks are not deleted together, orphan data may appear.
Insufficient validation of task assignee	Medium	A task may point to a non-existent user.
Inaccurate statistical scope	Low	Aggregation queries may produce biased results if filtering conditions are missing.

5.3 Testability Risks

Risk Item	Level	Description
Requirement description and implementation granularity are inconsistent	Medium	Some requirements are abstract, so additional branch information is needed during test design.
Read-only statistical requirements are not suitable for all black-box techniques	Low	For requirements with few input fields, EP and BVA outputs may be limited.
White-box coverage depends on the accuracy of requirement-code mapping	Medium	If the mapping range is too large, the coverage result may become low and hard to interpret.

6 Test Arrangement Based on Risks

6.1 Test Priority Allocation

Priority	Requirements	Test Arrangement
High	REQ-001, REQ-002, REQ-003	Execute first and cross-check with multiple black-box and white-box methods.
Medium	REQ-004, REQ-005, REQ-006	Combine input validation and key path testing to cover main business rules.
Low	REQ-007	Focus on statistical correctness and abnormal paths to verify basic behavior.

6.2 Testing Resource Suggestions

1. The authentication module is a high-risk area and should receive complete black-box test design first.
2. The task management module is the detailed testing target of this project and should use both black-box and white-box methods.
3. Statistical requirements can reduce field-level black-box design and emphasize decision table and path verification.
4. For modules with known requirement deviation risks, a manual review step should be retained.

7 Conclusion

From the overall risk distribution, the user authentication module is the highest-risk area of this project and should be the first focus of testing effort. Project management and task assignment come next, with emphasis on data consistency and object relationships. Task statistics has lower risk, but its aggregation logic still needs to be verified.

The analysis in this report provides the basis for the later test plan and detailed test design. The test scope, testing technique selection, and execution focus in later documents are aligned with the risk levels and testing priorities in this report.